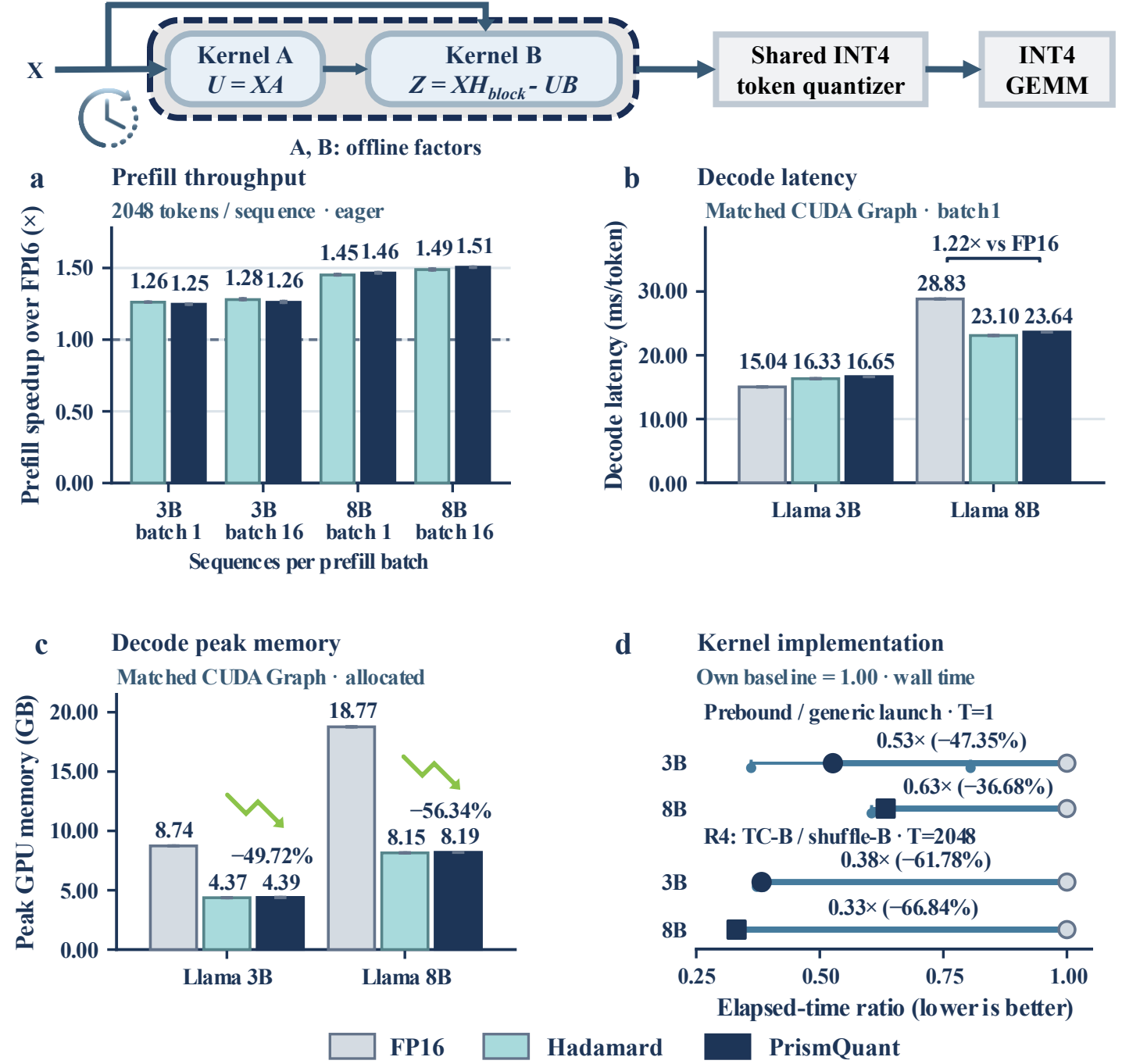

x
Kernel A
$U = XA$
Kernel B
$Z = XH_{block} - UB$
A, B: offline factors
Shared INT4
token quantizer
INT4
GEMM
a
Prefill throughput
2048 tokens / sequence · eager
Prefill speedup over FP16 (×)
1.50
1.00
0.50
0.00
1.26 1.25
1.28 1.26
1.45 1.46
1.49 1.51
3B batch 1
3B batch 16
8B batch 1
8B batch 16
Sequences per prefill batch
b
Decode latency
Matched CUDA Graph · batch1
Decode latency (ms/token)
30.00
20.00
10.00
0.00
1.22× vs FP16
28.83
23.10 23.64
15.04 16.33 16.65
Llama 3B
Llama 8B
c
Decode peak memory
Matched CUDA Graph · allocated
Peak GPU memory (GB)
20.00
15.00
10.00
5.00
0.00
8.74
4.37 4.39
−49.72%
18.77
8.15 8.19
−56.34%
Llama 3B
Llama 8B
d
Kernel implementation
Own baseline = 1.00 · wall time
Prebound / generic launch · T=1
3B
0.53× (−47.35%)
8B
0.63× (−36.68%)
R4: TC-B / shuffle-B · T=2048
3B
0.38× (−61.78%)
8B
0.33× (−66.84%)
0.25
0.50
0.75
1.00
Elapsed-time ratio (lower is better)
FP16
Hadamard
PrismQuant